DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTERSTUDIES

COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS
Second Semester, SY 2024-2025

COURSE CODE	**:**	S-CSGD324
COURSE TITLE	**:**	Immersive Game Development
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Two Units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	S-CSGD323LC
CO-REQUISITES	**:**	None
PROFESSOR	**:**	Mr. Jan Daenell Bancud
CONSULTATION HOURS	**:**	W/1000-1100/COS100-C
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VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course provides an in-depth exploration of game development using Unity, equipping students with the skills to design, prototype, and build interactive experiences. Students will learn industry-standard software design principles, apply the software development lifecycle in a game development context, and develop structured Game Design Documents (GDDs). Through hands-on projects, they will implement game mechanics, optimize performance, and create engaging 2D and 3D prototypes. The course emphasizes iteration, teamwork, and problem-solving, preparing students to design immersive worlds while upholding ethical and professional standards. By the end of the semester, students will have developed a working game prototype that showcases their creativity and technical expertise.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. describe and identify the core principles of immersive game world design while being capable of explaining foundational game design concepts;

CLO2. demonstrate understanding of game logic, mechanics and gameplay design;

CLO3. breakdown and categorize game ideas while analyzing feasibility of mechanics, illustrate and blueprint a game prototype;

CLO4. determine and evaluate the use of animations, timelines and UI elements for game design;

CLO5. assess and interpret how games can address social and environmental issues, evaluating existing games and creating new games that impact culture, inclusivity, and ethical considerations; and

CLO6. design, develop, and synthesize all learned concepts into a fully functional game prototype.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Describe and identify the core principles of immersive game world design while being capable of explaining foundational game design concepts.	**TLO1.** Define and differentiate game development and game design, while showcasing the importance of each.
	TLO2. Understand and learn the different tools, languages, approaches to game design, while providing knowledge about different game engines and platforms
CLO2. Demonstrate understanding of game logic, mechanics and gameplay design.	**TLO3.** Write modular, efficient scripts in C# to create objects, use functions and control game mechanics.
	TLO4. Employ concepts, unique ideas and novel mechanics into clear and easily to understand documentation
CLO3. Breakdown and categorize game ideas while analyzing feasibility of mechanics, illustrate and blueprint a game prototype	**TLO5.** Improve gameplay, consider debugging and iterative, continually improving releases from conception to finish.
CLO4. Determine and evaluate the use of animations, timelines and UI elements for game design.	**TLO6.** Determine how to incorporate animations, sounds, music and other elements to improve immersion and feel of games.

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 0	Week 1 Jan 20-25	• Gospel Reading, Reflection, and Prayer • Discussion of Course Syllabus • Course Requirements • Enabling Assessment Homework 1. - Unique Game Mechanics **Module 0: Course Introduction**	Schoolbook	✓		2
Module 1 CLO1 TLO1	Week 2 Jan 27-Feb 1	• Gospel Reading, Reflection, and Prayer • Introduction to Game Design **Module 1 - Introduction to Game Design**	Schoolbook	✓		2
Module 1 CLO1 TLO1	Week 3 Feb 3 - 8	• Gospel Reading, Reflection, and Prayer • Explore the Elements of Engaging Games	Schoolbook	✓		2

Module 2 CLO2 TLO2	Week 4 Feb 10-15	• Gospel Reading, Reflection, and Prayer • Game Design and Its significance • History of Video Games **Module 2 - Game Design Foundations**	Schoolbook	✓		2
Module 2 CLO2 TLO2	Week 5 Feb 17-22	• Gospel Reading, Reflection, and Prayer • Common Game Engines and Languages	Schoolbook		✓	2
Module 2 CLO2 TLO2	Week 6 Feb 24-Mar 1	• Gospel Reading, Reflection, and Prayer • 7 Phases of Game Development	Schoolbook	✓		2
Module 3 CLO3 TLO5	Week 7 Mar 3-8	• Gospel Reading, Reflection, and Prayer • Implementing Game Mechanics • How to debug • Enabling Assessment 2. Short Quiz - Game Design Fundamentals **Module 3 - Introduction to Game Development**	Schoolbook	✓		2

Module 3 CLO3 TLO5	Week 8 Mar 10-15	• Gospel Reading, Reflection, and Prayer • The game loop • Building and exporting your game • Enabling Assessment 3. Short Quiz - The Unity Editor	Schoolbook		✓	2
Midterm Exam	Week 9 Mar 17-22	• Gospel Reading, Reflection, and Prayer • **Midterm Summative Assessment on Modules 1-3**		✓		2
TOTAL				14	4	18

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO5. Assess and interpret how games can address social and environmental issues, evaluating existing games and creating new games that impact on culture, inclusivity, and ethical considerations.	**TLO7.** Incorporate good storytelling, immersive world building or deep game design elements to highlight social, environmental and economic concerns while presenting different perspectives and takes.
	TLO8. Demonstrate collaboration, teamwork and camaraderie by supporting and evaluating each other's works, providing a sense of unity and fulfillment among peers.
	TLO9. Create a reflective, retrospective analysis of the entire game design and development process for further learning.
CLO6. Design, develop, and synthesize all learned concepts into a fully functional game prototype.	**TLO10.** Develop a solid understanding of how to navigate and use the Unity Editor, as well as how to import and export objects, prefabs and components.
	TLO11. Create and design a working prototype/model of the game loop as well as the mechanics to incorporate.

	TLO12. Design and export the working game prototype for feedback and analysis

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 4 CLO5 TLO7	Week 10 Mar 24-29	• Gospel Reading, Reflection, and Prayer • Incorporating Storytelling/Worldbuilding • Incorporating Conflict • **Completion of Missed Assessment** • Enabling Assessment Homework 1. - Immersive Game Story **Module 4 - Immersive Game Design**	Schoolbook	✓		2
Module 4 CLO5 TLO7	Week 11 Mar 31-Apr 5	• Gospel Reading, Reflection, and Prayer • Creating Immersive environments • Enabling Assessment 2. Short Quiz - On Module 4	Schoolbook		✓	2
Module 5 CLO6 TL10 TL11 TL12	Week 12 Apr 7-12	• Gospel Reading, Reflection, and Prayer • Level design in Unity • Class Participation 1. Minigame Creation **Module 5 - Prefabs and Level Design**	Schoolbook	✓		2

	Week 13 Apr 14-19	***Independent Learning Days (April 14-15)*** ***Holy Week (April 16-19)***			✓	2
	Week 14 Apr 21-26	***Research Week***			✓	2
Module 5 CLO6 TL10 TL11 TL12	Week 15 Apr 28-May 3	• Gospel Reading, Reflection, and Prayer • Exporting scenes/levels • Exporting game assets for reuse • Enabling Assessment 3. Short Quiz - On Module 5	Schoolbook		✓	2
Module 6 CLO5 TLO9	Week 16 May 5-10	• Gospel Reading, Reflection, and Prayer • Creating a game development journal • Writing a post-mortem • Class Participation 2. Discussing game development journal entries with other students **Module 6 - Improving the Game Design Process**	Schoolbook	✓		2
Module 6 CLO5 TLO9	Week 17 May 12-17	• Gospel Reading, Reflection, and Prayer			✓	2

		• Project Development and Progress Report				
Final Exam	Week 18 May 19-24	• Gospel Reading, Reflection, and Prayer • **Final Term Summative Assessment (Presentations of Final Output – Working Game Prototype)**		✓		2
TOTAL				8	10	18

FINAL PRODUCT/PERFORMANCE:

At the end of the course, the students will collaboratively design and develop an immersive video game prototype that will incorporate all the game design principles, workflows, and elements learned from this course. The game should employ the following immersive game design principles:

- Mechanics/Gameplay – The game should clearly portray unique, novel or improved game mechanics from other games.
- Storytelling/World Building – The prototype should include a narrative, conflict and worldbuilding/lore aspect.
- Player motivation/fun – motivations that will keep the game interesting and the player invested in playing.
- Aesthetics – The prototype may or may not have impressive aesthetics, as long as the game prototype is playable.

Students will also work as a team and other teams to receive feedback and analyze the game design process. The students shall present their games in a public presentation-like fashion in which they will be able to employ their presentational, communicative and personal skills to market and showcase their game prototype.

RUBRIC FOR EVALUATING THE FINAL PRODUCT:

Criteria	Percentage
Creativity, Uniqueness and Design	25%
Documentation and Presentation	25%
Functionality and Fun	25%
Collaboration and Communication	15%
Group Impact/Feedback	10%

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
   a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
   b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examinations.
d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

Online References

Provider	Reference Material
Online	Unity Technologies. 2024. Game development resources, case studies & articles. Unity. Accessed February 8, 2025. **Available from:** https://unity.com/resources
Online	Gazis A, Katsiri E. 2023. Serious games in digital gaming: a comprehensive review of applications, game engines and advancements. arXiv. Accessed February 8, 2025. **Available from:** https://arxiv.org/abs/2311.03384
Online	Holly M, Habich L, Pirker J. 2024. GameDevDojo: an educational game for teaching game development concepts. arXiv. Accessed February 8, 2025. **Available from:** https://arxiv.org/abs/2407.12050
Online	University of Michigan Library. 2024. Databases and journals - video game development. Accessed February 8, 2025. **Available From:** **https://guides.lib.umich.edu/c.php?g=282990&p=1885548**
Online	Game Developer. 2024. Game industry news, deep dives, and analysis. Accessed February 8, 2025. **Available From: https://www.gamedeveloper.com/**
Online	Game Maker's Toolkit. 2025. Game design analysis and development resources. YouTube. Accessed February 8, 2025**.** **Available From: https://www.youtube.com/c/MarkBrownGMT**
Online	Extra Credits. 2025. Educational video essays on game design, history, and storytelling. YouTube. Accessed February 8, 2025 **Available From: https://www.youtube.com/extracredits/videos**
Online	Brackeys. 2025. Unity tutorials and game development education. YouTube. Accessed February 8, 2025. **Available from:** https://www.youtube.com/c/Brackeys

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2nd Semester 2024-2025

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